HAPPIEST HEALTH DIGITAL TWIN CHALLENGE · EXPERT JURY DEFENSE

T2D Glucose Digital Twin

A physiology-informed digital twin forecasting 2-hour glucose trajectories and stating exactly what its evidence does and does not prove.

Team: Akshaya R (BGSIT) & Nikhil VS (EWIT) · Bangalore, Karnataka
Scope: Research Prototype Only · Not for Clinical Treatment Decisions

The Clinical Problem: Reactive Alarms vs. Actionable Windows

Why postprandial glucose prediction matters in type 2 diabetes, and why it is not medical advice.

Current Reality: Reactive Threshold Alarms

- CGMs measure interstitial glucose every 5 minutes (5–15 min lag behind blood).
- Commercial alarms beep only AFTER glucose crosses 180 mg/dL.
- By then, carbohydrate absorption is in full progress.
- The practical window for behavioral intervention (e.g. a 15-min walk) has already closed.
- Patients experience alert fatigue from reactive alarms with high false-alert rates.

Our Objective: 30–60 Minute Advance Forecast

- Forecast continuous trajectory 120 minutes ahead with confidence intervals.
- Alert 30–60 minutes BEFORE crossing 180 mg/dL to enable non-drug action.
- Exercise activates GLUT4 muscle glucose uptake before the peak occurs.
- Strict False-Alert Budget: Keep false alarms ≤ 1.0 per patient-day to prevent fatigue.
- Crucial Distinction: Predictions are conditional estimates, NOT clinical prescriptions.

Our Solution: A Physiology-Informed Digital Twin Workstation

Coupling biophysical ODEs with attention transformers behind a production clinical UI.

1. Physiology Twin

- Modified Bergman Minimal ODEs model glucose clearance and gut absorption.
- Unscented Kalman Filter (UKF) recursively updates insulin action & sensitivity.
- Generates smooth, biologically plausible 2-hour mechanistic baseline.

2. Learned PR-TFT

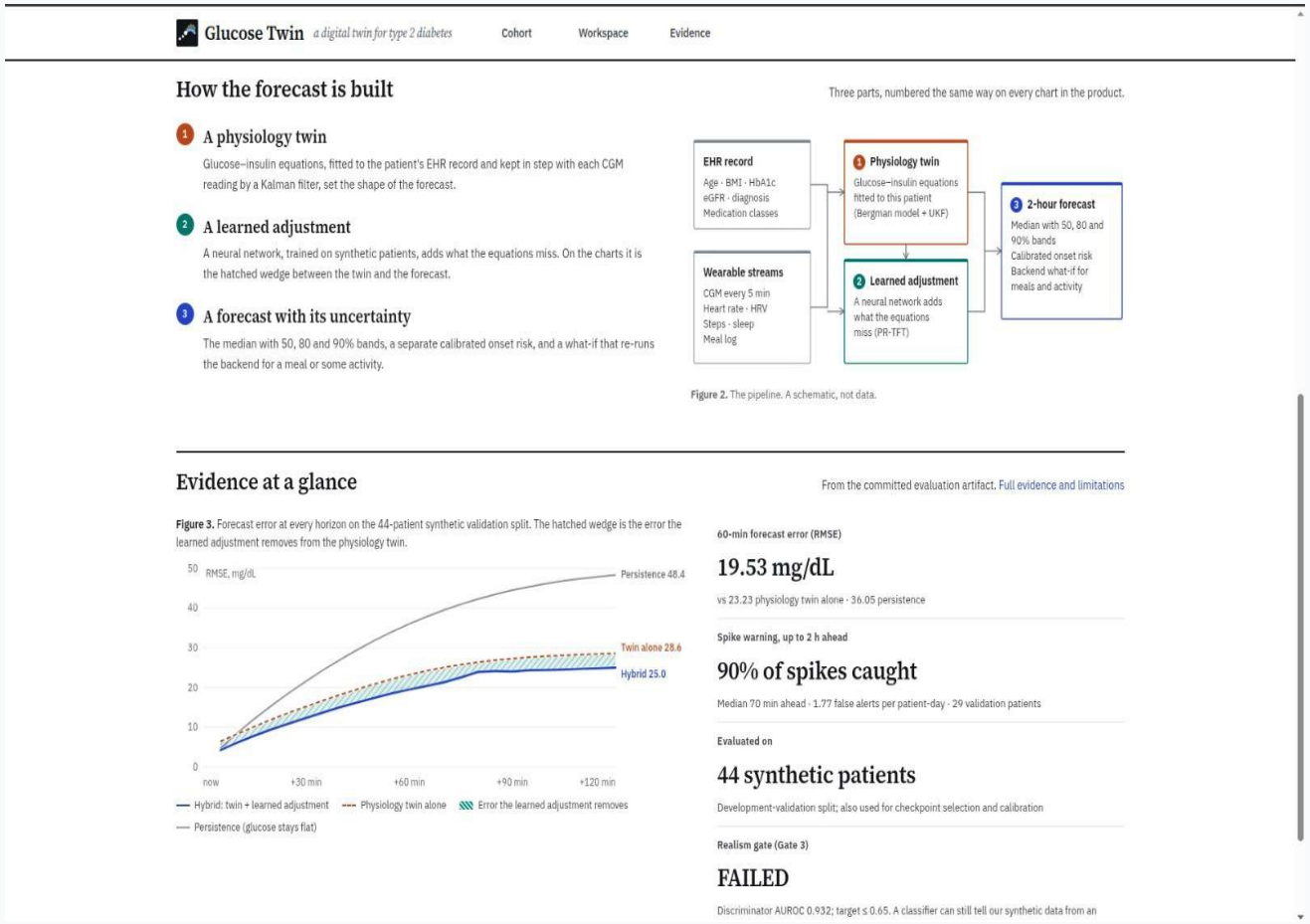
- Temporal Fusion Transformer learns strictly the residual the physics misses.
- Consumes EHR context, 24-hr wearable history (HR, steps, sleep), and clock.
- Predicts 7 quantiles (P05–P95) with conformal coverage & calibrated hazard.

3. Clinical Workstation

- Next.js 15 UI with Ward Triage and time-scrubbing replay navigation.
- What-If counterfactual scenario re-runs (meals, walking; refuses drugs).
- Bilingual patient note (English/Hindi) and official HL7 FHIR R4 export.

How It Works: Causal Architecture & Data Flow

Every forecast is an explicit sum: Twin Baseline + Learned Residual.

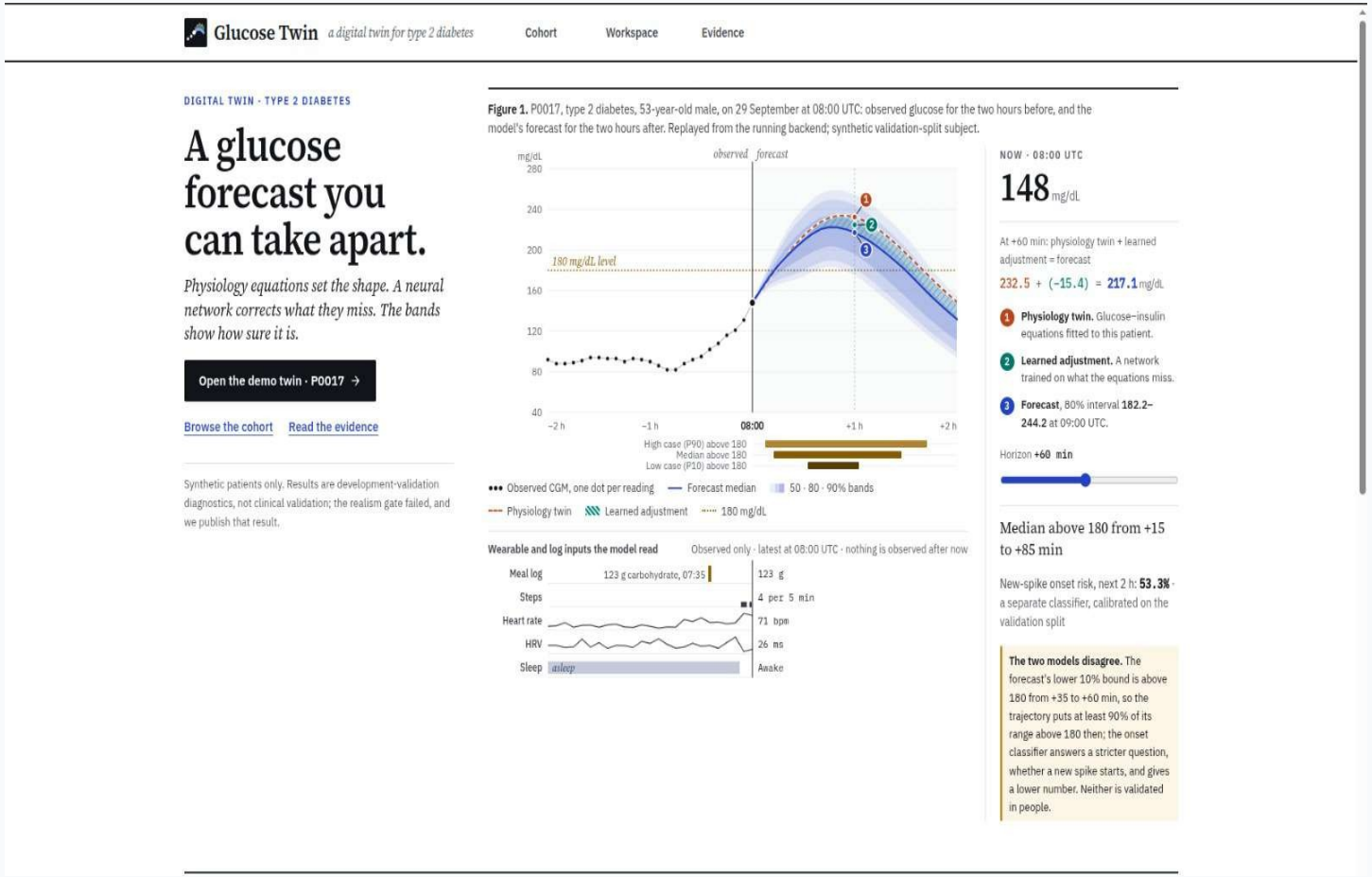


Mathematical Pipeline

1. Input Ingestion: 5-min CGM + static EHR (Age, BMI, HbA1c, eGFR) + wearables.
2. UKF Twin Rollout: Solves Bergman ODEs to produce mechanistic trajectory baseline.
3. Residual Attention: PR-TFT predicts delta ΔG across 7 quantiles (P05–P95).
4. Additive Composition: Final Forecast = Twin Rollout + Learned Adjustment.
5. Conformal Bounds: Empirical non-conformity guarantees 80% validation coverage.
6. Fallback Protection: If ML fails, system falls back to twin-only rollout.

Live Workstation: Verified Prototype on Synthetic Cohort

Real-time streaming replay of 249 served subjects; test partition strictly sealed.



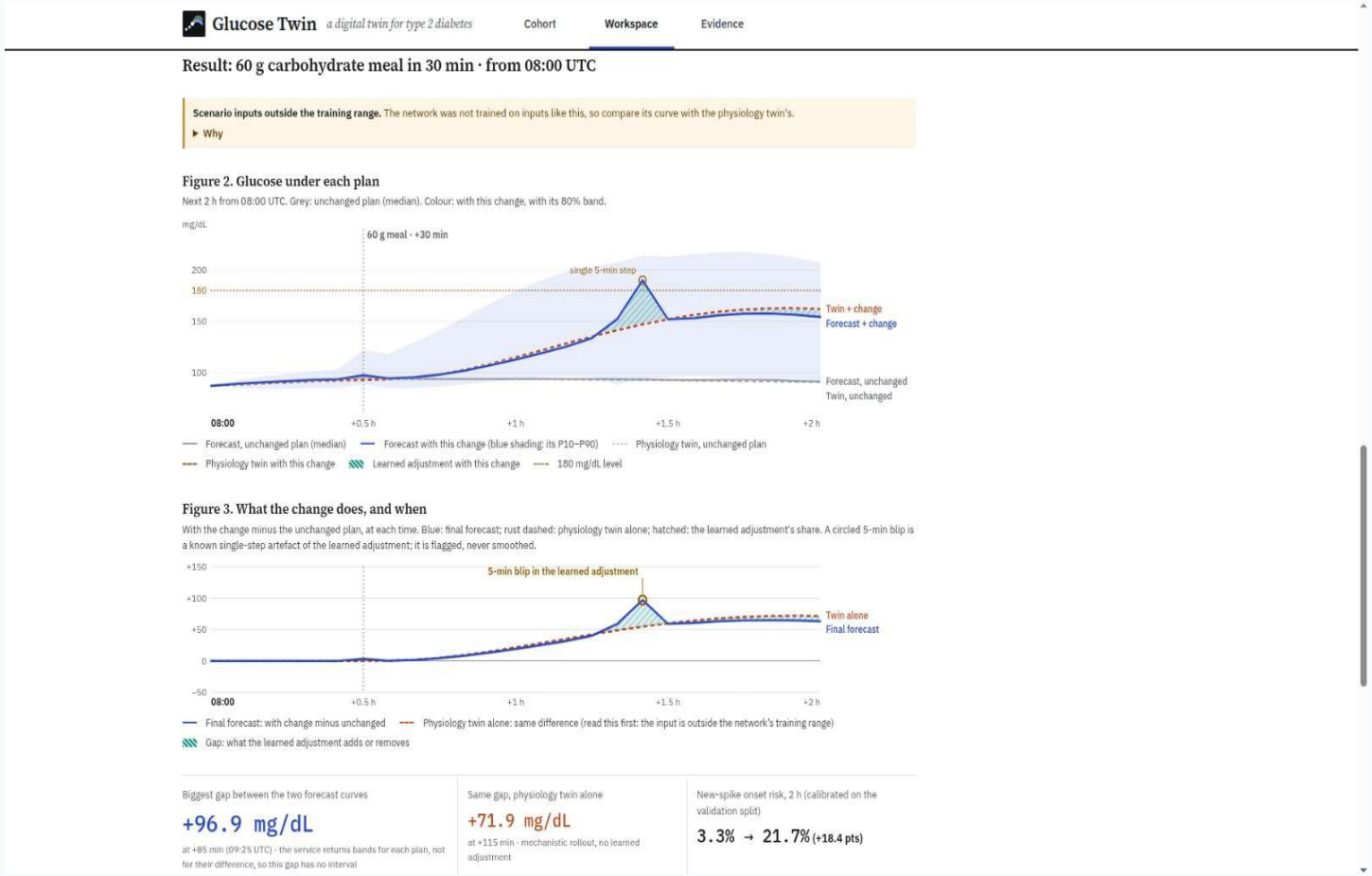
Verified Features

- Live Hero (P0017): 53yo male with T2D replayed at 08:00 UTC.
- On-Screen Math: Twin (232.5) + Adjustment (−15.4) = 217.1 mg/dL at +60m.
- Ward Triage View: Ranks patients by 2-hour spike-onset hazard.
- Timeline Pinning: Freeze historical moments without lookahead bias.
- Sealed Partition Guard: 44 test subjects return HTTP 404 (never exposed).

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Uncertainty & What-If Counterfactual Simulation

Conformal intervals, calibrated hazard, and principled scenario exploration.



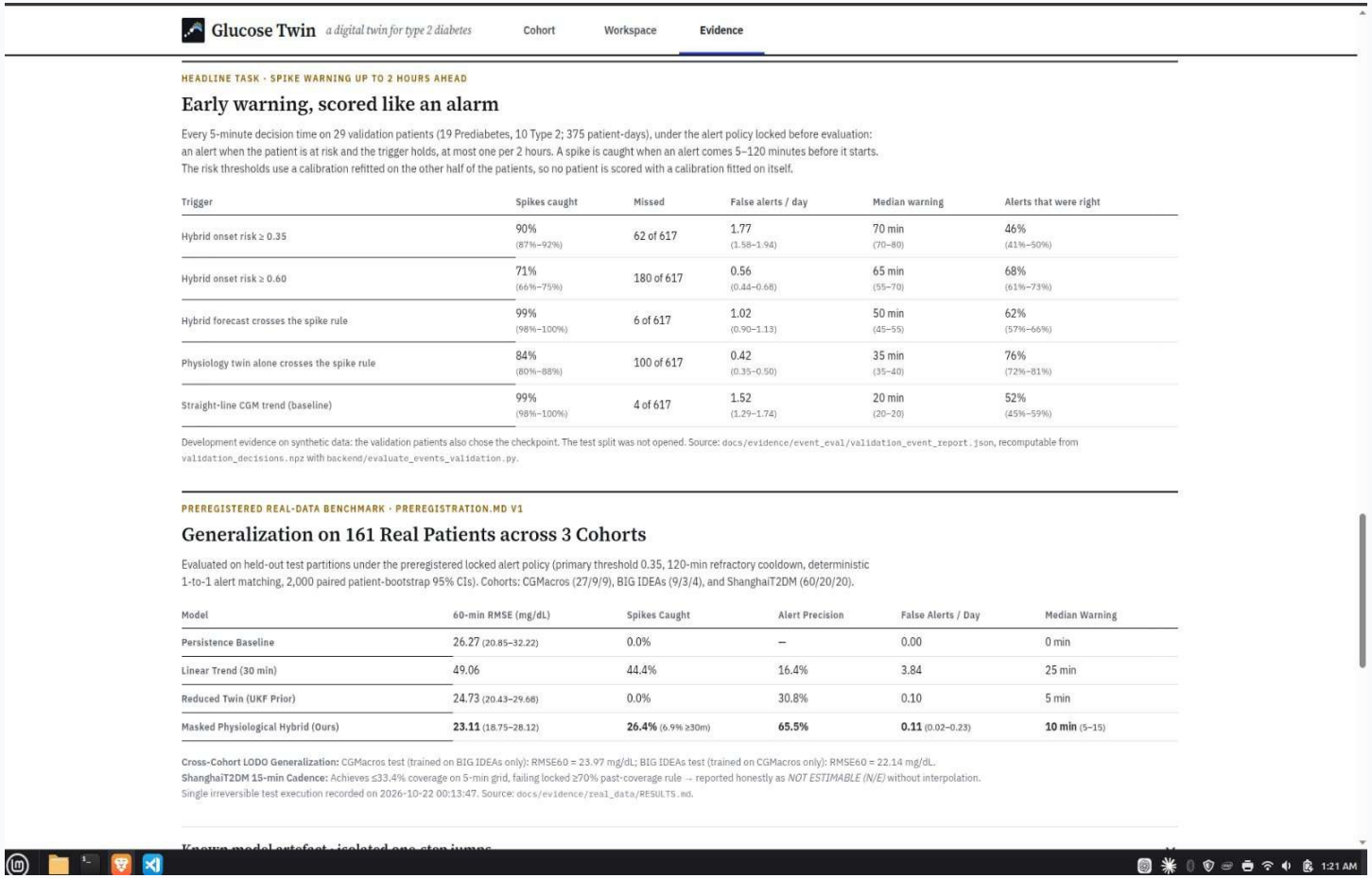
What-If Dynamics

- Add 60g Carbs in 30m: Mechanistic gut rollout raises glucose; network adjusts.
- Add 30m Walking: Elevates glucose clearance via exercise uptake.
- Medication Refusal: Clicking drug changes returns SCENARIO_UNSUPPORTED.
- Visible Artefact Flag: Circled 5-min blip is flagged, never silently smoothed.
- Causal Caveat: What-If is a simulated response, not a proven clinical outcome.

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Evidence: Locked Evaluation on 161 Real Patients

Primary locked test results on 11 held-out test patients with 5-minute CGM.



Glucose Twin *a digital twin for type 2 diabetes* Cohort Workspace Evidence

HEADLINE TASK · SPIKE WARNING UP TO 2 HOURS AHEAD

Early warning, scored like an alarm

Every 5-minute decision time on 29 validation patients (19 Prediabetes, 10 Type 2; 375 patient-days), under the alert policy locked before evaluation: an alert when the patient is at risk and the trigger holds, at most one per 2 hours. A spike is caught when an alert comes 5–120 minutes before it starts. The risk thresholds use a calibration refitted on the other half of the patients, so no patient is scored with a calibration fitted on itself.

Trigger	Spikes caught	Missed	False alerts / day	Median warning	Alerts that were right
Hybrid onset risk ≥ 0.35	90% (87%–92%)	62 of 617	1.77 (1.58–1.94)	70 min (70–80)	46% (41%–50%)
Hybrid onset risk ≥ 0.60	71% (66%–75%)	180 of 617	0.56 (0.44–0.68)	65 min (55–70)	68% (61%–73%)
Hybrid forecast crosses the spike rule	99% (98%–100%)	6 of 617	1.02 (0.90–1.13)	50 min (45–55)	62% (57%–66%)
Physiology twin alone crosses the spike rule	84% (80%–88%)	100 of 617	0.42 (0.35–0.50)	35 min (35–40)	76% (72%–81%)
Straight-line CGM trend (baseline)	99% (98%–100%)	4 of 617	1.52 (1.29–1.74)	20 min (20–20)	52% (45%–59%)

Development evidence on synthetic data: the validation patients also chose the checkpoint. The test split was not opened. Source: `docs/evidence/event_eval/validation_event_report.json`, recomputable from `validation_decisions.npz` with `backend/evaluate_events_validation.py`.

PREREGISTERED REAL-DATA BENCHMARK · PREREGISTRATION.MD V1

Generalization on 161 Real Patients across 3 Cohorts

Evaluated on held-out test partitions under the preregistered locked alert policy (primary threshold 0.35, 120-min refractory cooldown, deterministic 1-to-1 alert matching, 2,000 paired patient-bootstrap 95% CIs). Cohorts: CGMacros (27/9/9), BIG IDEAs (9/3/4), and ShanghaiT2DM (60/20/20).

Model	60-min RMSE (mg/dL)	Spikes Caught	Alert Precision	False Alerts / Day	Median Warning
Persistence Baseline	26.27 (20.85–32.22)	0.0%	–	0.00	0 min
Linear Trend (30 min)	49.06	44.4%	16.4%	3.84	25 min
Reduced Twin (UKF Prior)	24.73 (20.43–29.68)	0.0%	30.8%	0.10	5 min
Masked Physiological Hybrid (Ours)	**23.11** (18.75–28.12)	**26.4%** (6.9% ≥30m)	**65.5%**	**0.11** (0.02–0.23)	**10 min** (5–15)

Cross-Cohort LODO Generalization: CGMacros test (trained on BIG IDEAs only): RMSE60 = 23.97 mg/dL; BIG IDEAs test (trained on CGMacros only): RMSE60 = 22.14 mg/dL.
ShanghaiT2DM 15-min Cadence: Achieves ≤33.4% coverage on 5-min grid, failing locked ≥70% past-coverage rule → reported honestly as *NOT ESTIMABLE (N/E)* without interpolation.
Single irreversible test execution recorded on 2026-10-22 00:13:47. Source: `docs/evidence/real_data/RESULTS.md`.

Known model artefact - isolated one-step jumps

1:21 AM

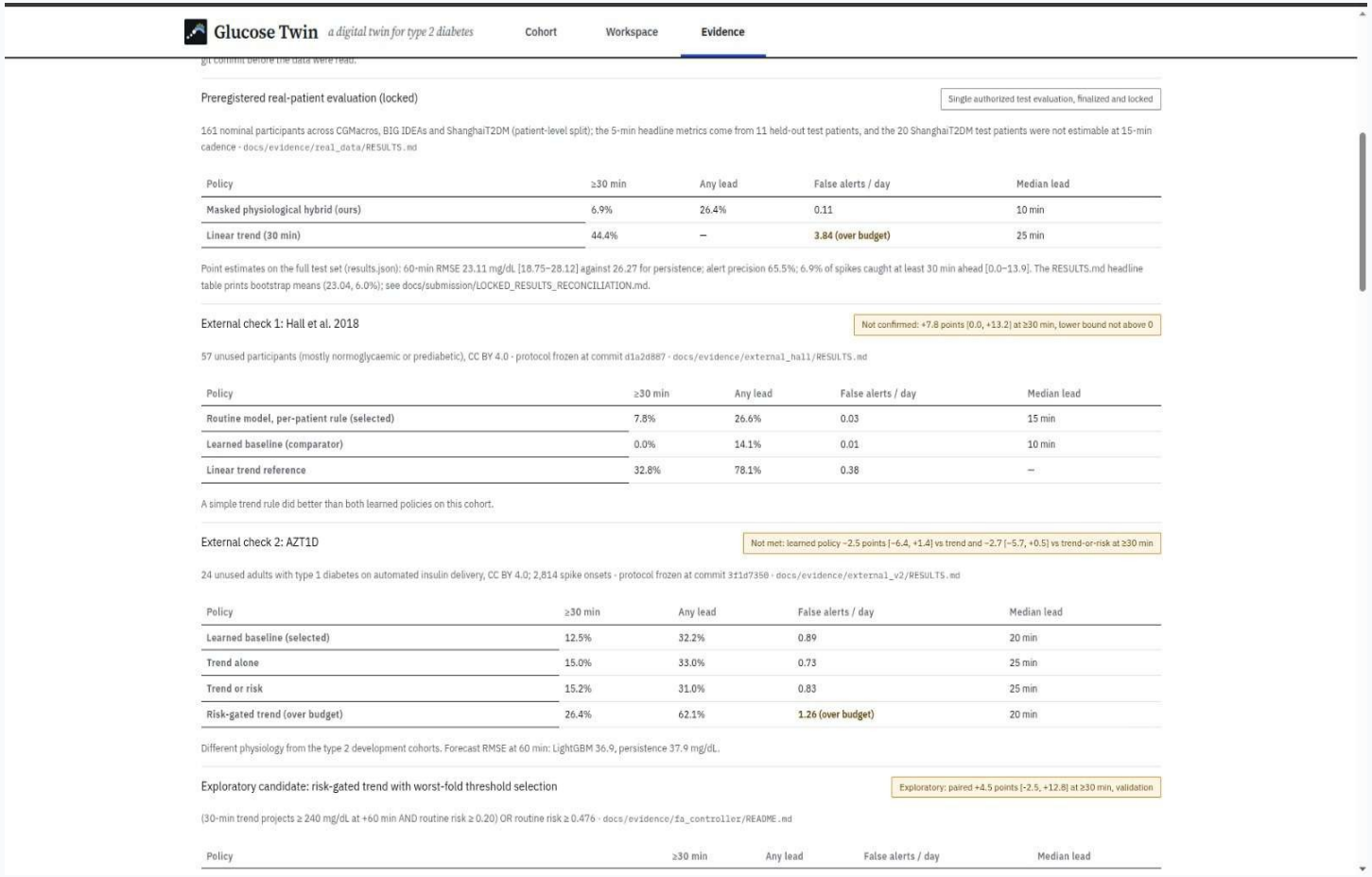
Key Verified Numbers

- 60-min RMSE: 23.11 mg/dL [18.75–28.12] (Persistence: 26.27 mg/dL).
- False Alerts: 0.11 / patient-day (within ≤ 1.0 budget).
- Alert Precision: 65.5% (Linear trend: only 16.4%).
- Early Sensitivity (≥30m): 6.9% point est. / 6.0% bootstrap mean.
- ShanghaiT2DM: Declared NOT ESTIMABLE (15-min cadence violates 70% rule).

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What Failed & What We Learned: Honest Science

Negative results, failed gates, and external validation checks.



Glucose Twin *a digital twin for type 2 diabetes* Cohort Workspace Evidence

git commit before the data were read.

Preregistered real-patient evaluation (locked) — Single authorized test evaluation, finalized and locked

161 nominal participants across CGMacros, BIG IDEAs and ShanghaiT2DM (patient-level split); the 5-min headline metrics come from 11 held-out test patients, and the 20 ShanghaiT2DM test patients were not estimable at 15-min cadence · docs/evidence/real_data/RESULTS.md

Policy	≥30 min	Any lead	False alerts / day	Median lead
Masked physiological hybrid (ours)	6.9%	26.4%	0.11	10 min
Linear trend (30 min)	44.4%	–	3.84 (over budget)	25 min

Point estimates on the full test set (results.json): 60-min RMSE 23.11 mg/dL [18.75–28.12] against 26.27 for persistence; alert precision 65.5%; 6.9% of spikes caught at least 30 min ahead [0.0–13.9]. The RESULTS.md headline table prints bootstrap means (23.04, 6.0%); see docs/submission/LOCKED_RESULTS_RECONCILIATION.md.

External check 1: Hall et al. 2018 — Not confirmed: +7.8 points [0.0, +13.2] at ≥30 min, lower bound not above 0

57 unused participants (mostly normoglycaemic or prediabetic), CC BY 4.0 · protocol frozen at commit d1a2d887 · docs/evidence/external_hall/RESULTS.md

Policy	≥30 min	Any lead	False alerts / day	Median lead
Routine model, per-patient rule (selected)	7.8%	26.6%	0.03	15 min
Learned baseline (comparator)	0.0%	14.1%	0.01	10 min
Linear trend reference	32.8%	78.1%	0.38	–

A simple trend rule did better than both learned policies on this cohort.

External check 2: AZT1D — Not met: learned policy −2.5 points [−6.4, +1.4] vs trend and −2.7 [−5.7, +0.5] vs trend-or-risk at ≥30 min

24 unused adults with type 1 diabetes on automated insulin delivery, CC BY 4.0; 2,814 spike onsets · protocol frozen at commit 3f1d7350 · docs/evidence/external_v2/RESULTS.md

Policy	≥30 min	Any lead	False alerts / day	Median lead
Learned baseline (selected)	12.5%	32.2%	0.89	20 min
Trend alone	15.0%	33.0%	0.73	25 min
Trend or risk	15.2%	31.0%	0.83	25 min
Risk-gated trend (over budget)	26.4%	62.1%	1.26 (over budget)	20 min

Different physiology from the type 2 development cohorts. Forecast RMSE at 60 min: LightGBM 36.9, persistence 37.9 mg/dL.

Exploratory candidate: risk-gated trend with worst-fold threshold selection — Exploratory: paired +4.5 points [-2.5, +12.8] at ≥30 min, validation

(30-min trend projects ≥ 240 mg/dL at +60 min AND routine risk ≥ 0.20) OR routine risk ≥ 0.476 · docs/evidence/fa_controller/README.md

Policy	≥30 min	Any lead	False alerts / day	Median lead

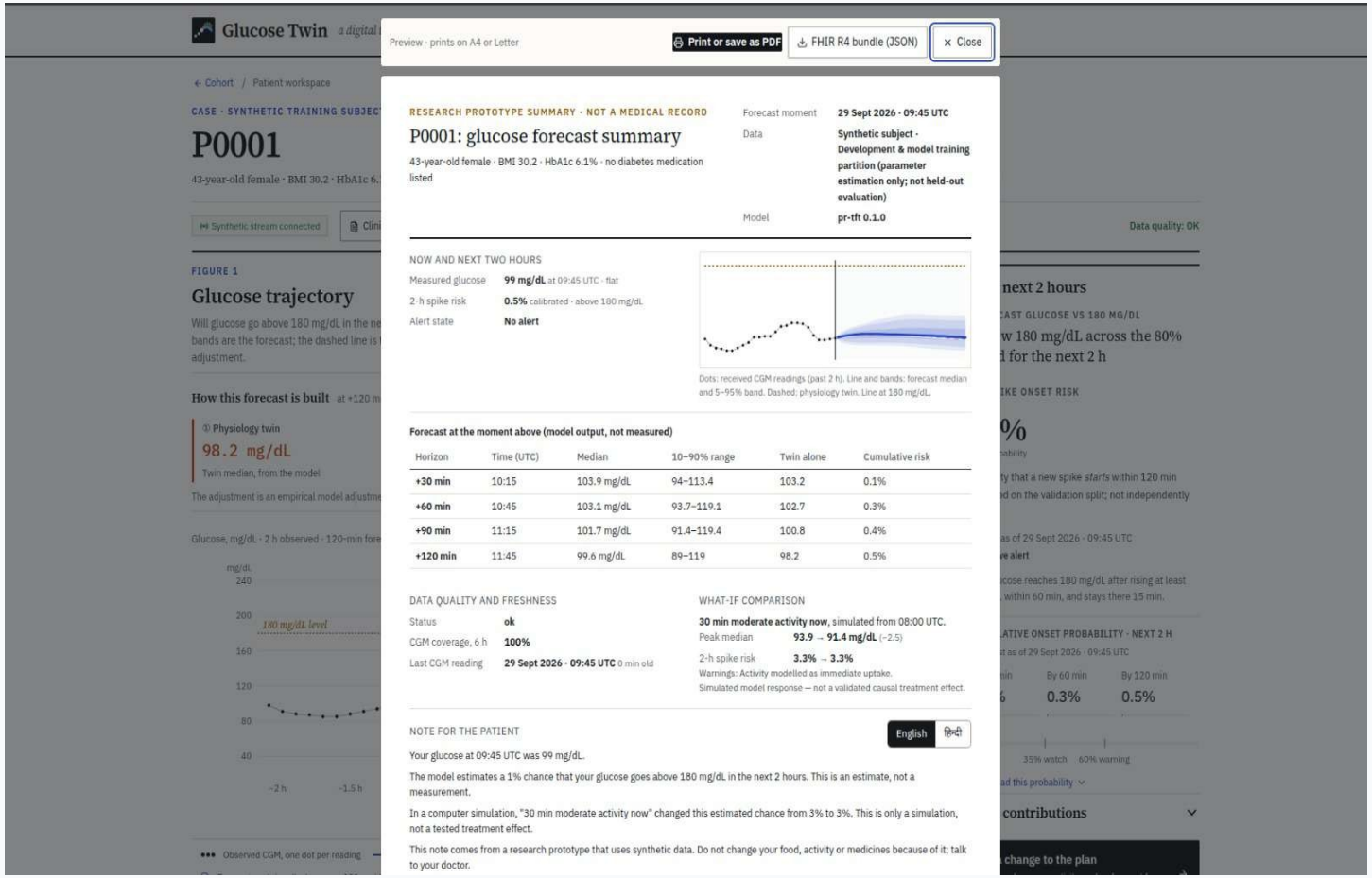
Failed Gates Disclosed

- Gate 3 Realism Audit: FAILED (AUROC 0.932 vs ≤0.65 target; ODE traces too smooth).
- Real Early Warning (6.9%): Low advance sensitivity due to unannounced meals.
- External Check 1 (Hall 2018): NOT CONFIRMED (+7.8%, 95% CI crosses 0).
- External Check 2 (AZT1D): NOT MET (Learned model −2.5% vs linear trend).
- Lesson: Probability thresholds do not transfer across cohorts without local recalibration.

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Engineering Rigor, Safety Governance & Interoperability

Test split firewalls, HL7 FHIR R4 export, and containerized deployment.



Clinical Integration

- Partition Firewall: Sealed test partition isolated at API boundary (HTTP 404).
- HL7 FHIR R4: Validated JSON export using LOINC 99504-3 and HTEST security tag.
- Bilingual Note: Plain English and Hindi patient summaries.
- Reproducibility: Git SHA commit tracking, deterministic seeds, and run manifests.
- Zero Real Data Leakage: Real patient datasets stored offline only.

Conclusion & Next Steps: The Path to Clinical Translation

What we proved, what remains, and our evidence-justified research roadmap.

What We Have Proven

✔️ A working, reproducible digital twin combining biophysical ODEs with PR-TFT attention.

✔️ A 16.0% RMSE reduction over the physics twin on synthetic validation data.

✔️ A 60-min RMSE of 23.11 mg/dL on held-out real patients within a 0.11 false alert/day budget.

✔️ Full-stack clinical workstation with FHIR export, What-If simulation, and sealed partition guards.

✔️ Uncompromising scientific integrity: failed gates and negative findings published openly.

Next Steps Before Clinical Use

➔ 1. AI-READI Confirmation: Execute frozen protocol on ~944 T2D participants once data access is approved.

➔ 2. Adaptive Thresholds: Implement dynamic operating points to prevent cross-cohort calibration collapse.

➔ 3. Unannounced Meal Deconvolution: Infer carbohydrate arrival directly from early CGM inflections.

➔ 4. Curvature Regularization: Retrain PR-TFT with smoothness loss to permanently eliminate 20 mg/dL jumps.

➔ 5. Observational Clinical Pilot: Evaluate clinician decision support in outpatient type 2 reviews.

Slide 11: Appendix — Bergman Minimal Model & UKF Equations

Technical reference & deep-dive material for jury questions.

- Glucose dynamics: $dG/dt = -[p1 + X(t)]\, G(t) + p1\, Gb + (1/VG)\, Ra(t)$
- Remote insulin action: $dX/dt = -p2\, X(t) + p3\, [I(t) - Ib]$
- Gut absorption: Two-compartment model $dQ1/dt = -k_gri\, Q1 + Meals$; $dQ2/dt = k_gri\, Q1 - k_abs\, Q2$
- UKF State Vector: [G, X, I, Q1, Q2, SI, EGP] tracked recursively with sigma points.
- State prior prevents unphysical negative glucose or runaway exponential spikes during dropouts.

Slide 12: Appendix — PR-TFT Architecture & Hyperparameters

Technical reference & deep-dive material for jury questions.

- Architecture: Temporal Fusion Transformer with Gated Residual Networks (GRN) and Variable Selection Networks (VSN).
- Hidden dimension: 96; Attention heads: 4; Layers: 2; Dropout: 0.1.
- Inputs: Static EHR (Age, BMI, HbA1c, eGFR) + Dynamic Wearables (CGM, HR, HRV, Steps, Sleep) + Clock features.
- Loss function: Pinball quantile loss across 7 quantiles (P05 to P95) + Binary cross-entropy on hazard head.
- Early stopping: Patience of 4 epochs on validation loss (checkpoint: epoch=03, val_loss=0.2255).

Slide 13: Appendix — Dataset Split Accountability & Denominators

Technical reference & deep-dive material for jury questions.

- Synthetic Cohort: 293 total (205 train, 44 val, 44 test sealed). Served in app: 249 subjects.
- Real Benchmark: 161 nominal patients across CGMacros (45), BIG IDEAs (16), ShanghaiT2DM (100).
- ShanghaiT2DM: 15-minute sampling cadence violates ≥70% 5-minute coverage rule -> NOT ESTIMABLE.
- Evaluated Real Test Set: Exactly 11 held-out test patients with native 5-minute CGM.
- Zero participant-level outcome inspection was performed during development.

Slide 14: Appendix — Metric Definitions & 1-to-1 Alert Matching

Technical reference & deep-dive material for jury questions.

- Adverse Spike: Crossing ≥180 mg/dL preceded by ≥50 mg/dL rise in 60 min, persisting for ≥15 min.
- 1-to-1 Alert Matching: An alert matches at most ONE subsequent spike event (5–120 min ahead).
- Refractory Cooldown: 120-minute alert suppression following any triggered alarm.
- Sensitivity ≥30m: Spikes caught with an alert at least 30 minutes before crossing 180 mg/dL.
- False Alert Budget: Maximum allowable rate is ≤ 1.0 false alert per patient-day.

Slide 15: Appendix — External Checks on Unused Cohorts (Hall & AZT1D)

Technical reference & deep-dive material for jury questions.

- Hall et al. 2018 (57 participants, 64 spikes): Routine model +7.8% [0.0, +13.2]; NOT CONFIRMED.
- Linear trend beat learned models on Hall (32.8% sens at 0.38 FA/day) due to low event frequency.
- AZT1D (24 adults with T1D on AID, 2,814 spikes): Learned model 12.5% vs Trend 15.0%; NOT MET.
- Risk-Gated Trend (C2): Doubled sensitivity (26.4%) but exceeded false alert budget (1.26/day).
- Core Takeaway: Probability calibration collapses across populations without adaptive operating points.

Slide 16: Appendix — Sealed Partition Firewall Verification

Technical reference & deep-dive material for jury questions.

- Code Location: backend/serve_pr_tft.py line 79 (SEALED_SPLITS = frozenset({'test'})).
- Rejection Behavior: Querying any test-split subject UUID or patient ID raises HTTP 404.
- Subject Directory: GET /v1/subjects filters out all test IDs at the database iterator level.
- Offline Isolation: Real patient files (data/real/) are excluded from Git and Docker images.
- No Lookahead Leakage: Timeline scrubbing strictly hides readings after the anchor timestamp.

Slide 17: Appendix — Clinical Safety, Medication Refusal & FHIR R4

Technical reference & deep-dive material for jury questions.

- Medication Refusal: What-If drug modifications return 422 SCENARIO_UNSUPPORTED.
- FHIR R4 Schema: Exported bundles validated against official HL7 JSON schema.
- Standard Codings: Interstitial glucose coded as LOINC 99504-3 with UCUM units mg/dL.
- HTEST Security Label: Prevents accidental ingestion into real clinical EHR databases.
- Bilingual Translation: English and Hindi patient summaries with explicit clinical disclaimer.

Slide 18: Appendix — Unexecuted Protocols & Research Integrity

Technical reference & deep-dive material for jury questions.

- EW-01 Protocol: Prospective early-warning trial protocol remains unexecuted and excluded from claims.
- AI-READI Confirmation: Frozen preregistration on FAIRhub (~944 T2D participants); awaiting data access.
- Frozen Harness: aireadi_evaluate.py committed and SHA-256 hashed before receiving data.
- No P-Hacking: All evaluations executed in a single authorized run without parameter retuning.
- Complete Provenance: Every claim backed by commit SHAs and locked manifest files.